21 - (MathAA-W 2021-Paper 2/0-Q5) **-** *SEQUENCES - SERIES*

[Maximum mark: 9]

The sum of the first n terms of a geometric sequence is given by $S_n = \sum_{r=1}^{n} \frac{2}{3}\left(\frac{7}{8}\right)^r$.

(a) Find the first term of the sequence, u_1. [2]

(b) Find S_∞. [3]

(c) Find the least value of n such that $S_\infty - S_n < 0.001$. [4]

1 - (MAT-W 2012-Paper 2/0-Q10) - *COMPLEX NUMBERS*

[Maximum mark: 7]

Let $\omega = \cos\theta + \mathrm{i}\sin\theta$. Find, in terms of θ, the modulus and argument of $(1-\omega^2)^*$.

2 - (MAT-S 2013-Paper 2/1-Q6) **-** *COMPLEX NUMBERS*

6. *[Maximum mark: 7]*

A polynomial $p(x)$ with real coefficients is of degree five. The equation $p(x)=0$ has a complex root $2+\mathrm{i}$. The graph of $y=p(x)$ has the x-axis as a tangent at $(2, 0)$ and intersects the coordinate axes at $(-1, 0)$ and $(0, 4)$.

Find $p(x)$ in factorised form with real coefficients.

3 – (MAT-W 2013-Paper 2/0-Q6) – *COMPLEX NUMBERS*

A complex number z is given by $z = \dfrac{a+\mathrm{i}}{a-\mathrm{i}}$, $a \in \mathbb{R}$.

(a) Determine the set of values of a such that

(i) z is real;

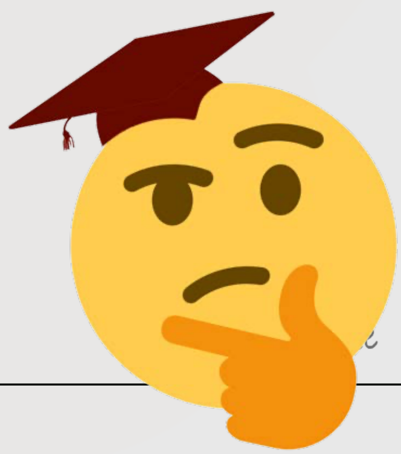
(ii) z is purely imaginary. *[4]*

(b) Show that $|z|$ is constant for all values of a. *[2]*

4 - (MAT-S 2014-Paper 2/1-Q1) **-** *COMPLEX NUMBERS*

[Maximum mark: 4]

One root of the equation $x^2 + ax + b = 0$ is $2 + 3\mathrm{i}$ where $a,\ b \in \mathbb{R}$. Find the value of a and the value of b.



5 - (MAT-S 2014-Paper 2/2-Q13) - *COMPLEX NUMBERS*

Consider $z = r(\cos\theta + \mathrm{i}\sin\theta)$, $z \in \mathbb{C}$.

(a) Use mathematical induction to prove that $z^n = r^n(\cos n\theta + \mathrm{i}\sin n\theta)$, $n \in \mathbb{Z}^+$. *[7]*

Given $u = 1 + \sqrt{3}\mathrm{i}$ and $v = 1 - \mathrm{i}$,

(b) (i) express u and v in modulus-argument form;

(ii) hence find u^3v^4. *[4]*

The complex numbers u and v are represented by point A and point B respectively on an Argand diagram.

(c) Plot point A and point B on the Argand diagram. *[1]*

Point A is rotated through $\frac{\pi}{2}$ in the anticlockwise direction about the origin O to become point A'. Point B is rotated through $\frac{\pi}{2}$ in the clockwise direction about O to become point B'.

(d) Find the area of triangle $\mathrm{OA}'\mathrm{B}'$. *[3]*

Given that u and v are roots of the equation $z^4 + bz^3 + cz^2 + dz + e = 0$, where $b, c, d, e \in \mathbb{R}$,

(e) find the values of b, c, d and e. *[5]*

6 - (MAT-S 2015-Paper 2/1-Q12) **-** *COMPLEX NUMBERS*

(a) (i) Use the binomial theorem to expand $(\cos\theta + \mathrm{i}\sin\theta)^5$.

(ii) Hence use De Moivre's theorem to prove

$$\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta.$$

(iii) State a similar expression for $\cos 5\theta$ in terms of $\cos\theta$ and $\sin\theta$. [6]

Let $z = r(\cos\alpha + \mathrm{i}\sin\alpha)$, where α is measured in degrees, be the solution of $z^5 - 1 = 0$ which has the smallest positive argument.

(b) Find the value of r and the value of α. [4]

(c) Using (a) (ii) and your answer from (b) show that $16\sin^4\alpha - 20\sin^2\alpha + 5 = 0$. [4]

(d) Hence express $\sin 72^\circ$ in the form $\dfrac{\sqrt{a + b\sqrt{c}}}{d}$ where $a, b, c, d \in \mathbb{Z}$. [5]

7 - (MAT-S 2016-Paper 2/1-Q9) **-** *COMPLEX NUMBERS*

Two distinct roots for the equation $z^4 - 10z^3 + az^2 + bz + 50 = 0$ are $c + \mathrm{i}$ and $2 + \mathrm{i}d$ where $a, b, c, d \in \mathbb{R}, d > 0$.

(a) Write down the other two roots in terms of c and d. [1]

(b) Find the value of c and the value of d. [6]

8 - (MAT-S 2018-Paper 2/2-Q1) **-** *COMPLEX NUMBERS*

Consider the complex number $z = \dfrac{2 + 7\mathrm{i}}{6 + 2\mathrm{i}}$.

(a) Express z in the form $a + \mathrm{i}b$, where $a, b \in \mathbb{Q}$. [2]

(b) Find the exact value of the modulus of z. [2]

(c) Find the argument of z, giving your answer to 4 decimal places. [2]

9 - (MAT-S 2019-Paper 2/1-Q2) **-** *COMPLEX NUMBERS*

Solve $z^2 = 4e^{\frac{\pi}{2}i}$, giving your answers in the form

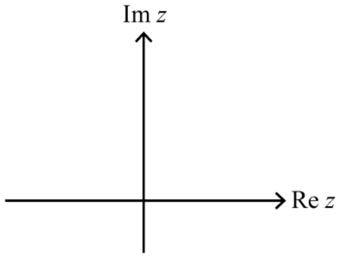
(a) $re^{i\theta}$ where $r, \theta \in \mathbb{R}, r > 0$; [3]

(b) $a + ib$ where $a, b \in \mathbb{R}$. [2]

10 - (MAT-S 2019-Paper 2/1-Q6) - *COMPLEX NUMBERS*

Let $z = a + bi$, $a, b \in \mathbb{R}^+$ and let $\arg z = \theta$.

(a) Show the points represented by z and $z - 2a$ on the following Argand diagram. [1]



(b) Find an expression in terms of θ for

(i) $\arg(z - 2a)$;

(ii) $\arg\left(\dfrac{z}{z-2a}\right)$. [3]

(c) Hence or otherwise find the value of θ for which $\mathrm{Re}\left(\dfrac{z}{z-2a}\right) = 0$. [3]

11 - (MAT-S 2019-Paper 2/2-Q8) - *COMPLEX NUMBERS*

(a) Find the roots of the equation $w^3 = 8\text{i}$, $w \in \mathbb{C}$. Give your answers in Cartesian form. [4]

One of the roots w_1 satisfies the condition $\text{Re}(w_1) = 0$.

(b) Given that $w_1 = \dfrac{z}{z - \text{i}}$, express z in the form $a + b\text{i}$ where $a, b \in \mathbb{Q}$. [3]

12 - (MAT-W 2019-Paper 2/0-Q6) - *COMPLEX NUMBERS*

[Maximum mark: 6]

Let $P(z) = az^3 - 37z^2 + 66z - 10$, where $z \in \mathbb{C}$ and $a \in \mathbb{Z}$.

One of the roots of $P(z) = 0$ is $3 + \mathrm{i}$. Find the value of a.

13 - (MathAA-S 2021-Paper 2/1-Q8) **-** *COMPLEX NUMBERS*

[Maximum mark: 7]

Consider the complex numbers $z = 2\left(\cos\frac{\pi}{5} + \mathrm{i}\sin\frac{\pi}{5}\right)$ and $w = 8\left(\cos\frac{2k\pi}{5} - \mathrm{i}\sin\frac{2k\pi}{5}\right)$, where $k \in \mathbb{Z}^+$.

(a) Find the modulus of zw. [1]

(b) Find the argument of zw in terms of k. [2]

Suppose that $zw \in \mathbb{Z}$.

(c) (i) Find the minimum value of k.

(ii) For the value of k found in part (i), find the value of zw. [4]

14 - (MathAA-S 2021-Paper 2/2-Q8) **-** *COMPLEX NUMBERS*

[Maximum mark: 5]

Consider $z = \cos\theta + \mathrm{i}\sin\theta$ where $z \in \mathbb{C}$, $z \neq 1$.

Show that $\mathrm{Re}\left(\dfrac{1+z}{1-z}\right) = 0$.